

CHRISTOPHER OOI EN SHEN

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Education

Wilfrid Laurier University

Sep. 2023 – May. 2027

Bachelor of Science in Computer Science

Waterloo, Ontario

- CGPA of **3.85**
- **Relevant Coursework:** Operating Systems, Data Structures & Algorithms, Object Oriented Programming, System Programming, Computer Networks, Database Systems

Experience

Hecaton 🌐

Oct. 2025 – Present

Contract Software Engineer

Waterloo, Ontario

- Delivered a production-ready **Bitcoin notarization platform** by engineering a secure **full-stack** architecture (**NextJs**, **Supabase**, **Prisma**, **bitcoinjs-lib**), successfully scaling to process **500+** monthly transactions post-launch.

Bolttech 🌐

May. 2025 – Sep. 2025

Data Engineer Intern

Kuala Lumpur, Malaysia

- Generated **\$15000+** in **annual savings** by **re-architecting legacy automation** into a **serverless AWS ETL pipeline (Lambda/Step Functions)** scaling to process **1M+** daily records.
- Engineered high-performance **Pandas** and **Databricks pipelines** to handle large-scale datasets, overcoming **memory constraints** and reduced data processing time by **95%** (**45m** → **2m**).

Sensoft Technologies 🌐

May. 2024 – Sep. 2024

Software Engineer Intern

Penang, Malaysia

- Built a **React** dashboard visualizing **15,000+** **Modbus sensors**, implementing **list virtualization** to maintain **60 FPS** rendering under high-frequency **WebSocket** load.
- **Reduced** energy costs by **15%** by developing a **hybrid analysis pipeline** utilizing **Facebook Prophet** for **time-series forecasting** and **OpenAI API** to generate automated actionable **reports**.
- Achieved **sub-100ms system latency** for **real-time people detection** by implementing a **multi-threaded** video pipeline to **parallelize frame capture** and **YOLOv8** inference tasks on **edge devices**.

Personal Projects

MentorTrader 🌐 | Next.js, PostgreSQL, Stripe, AWS, Sentry

- Created a **marketplace platform** focused on connecting **day traders with credible online trading mentors**.
- Onboarded **120+ active mentors** and **2,000+ monthly visitors**, featuring management **dashboards** and **partnerships** with companies like **Darwinex**, **Whop**.
- **Deployed** a **Next.js** application on **Vercel**, using **Supabase** for **secure authentication** and **database**, **Redis** for **caching**, **Sentry** for **observability**, and **Stripe** for **payments**.

Faves 🌐 | Express.js, Firebase, TailwindCSS, Google Cloud, OpenAI

- Built a discovery **platform** aggregating **1,000+ AI-verified recommendations** from influential figures.
- Features an **automated content pipeline** where public submissions trigger **LLM-based citation generation** followed by a rigorous **admin moderation** workflow.
- Architected a **serverless backend** using **Google Cloud Functions** and **Node.js**, securing public endpoints via **Joi** validation, **rate-limiting**, and **reCAPTCHA** to prevent abuse.

URL Shortener Website 🌐 | Ruby, HTML/CSS, JavaScript, PostgreSQL

- Engineered a high-performance link management service focusing on granular **traffic analytics** and **bot protection**.
- Captures **sub-millisecond telemetry** including visitor **geolocation** and device **fingerprinting**, while automatically enriching link metadata via background **scraping**.
- Built with **Ruby on Rails**, utilizing a **Write-Behind caching strategy** with **Redis** for instant redirects and **Sidekiq** to **offload heavy analytics writes** asynchronously.

Technical Skills

Languages: Python, C, JavaScript, TypeScript, HTML/CSS, Java, Ruby, VBA, SQL

Frameworks: ReactJs, NextJs, ExpressJs, FastApi, Flask, Rails, Jest

Developer Tools: AWS, Docker, Google Cloud Platform, Vercel, Postman, Heroku, npm, Redis, Git, Sentry, Github Actions, PostgreSQL, Firebase, Prisma